

Pavan Kumar

Bengaluru / Hyderabad (Open to relocate)

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Professional Summary

Data Scientist and GenAI Engineer with experience building retrieval-augmented generation (RAG) systems, agentic AI workflows, and machine learning solutions on Azure. Skilled in prompt engineering, structured outputs, evaluation pipelines, and scalable ML practices with a focus on business impact and responsible AI.

Technical Skills

Programming	Python, SQL, Git
Machine Learning	scikit-learn, XGBoost, LightGBM, Model Evaluation, Imbalanced Learning
GenAI / NLP	RAG, Embeddings, Prompt Engineering, Context Windows, Structured Outputs
Agentic AI	LangGraph, Tool Calling, Routing, ReAct Workflows, Memory Patterns
Frameworks	LangChain, LlamaIndex, FastAPI, Flask, Pydantic
Vector Search	Azure AI Search, FAISS, ChromaDB
Azure	Azure OpenAI, Azure ML, Blob Storage, ADLS, Key Vault
MLOps	MLflow, Offline Evaluation, Monitoring, Experiment Tracking
Data Stack	Pandas, NumPy, EDA, Feature Engineering, Data Validation

Experience

INTENSO TECH SOLUTION PRIVATE LIMITED

Data Scientist / GenAI Engineer

Bangalore, India

Mar 2024 - Mar 2026 (2+ Years)

- Developed and deployed AI and machine learning solutions for enterprise applications using Python and Azure-based technologies.
- Built retrieval-augmented generation (RAG) pipelines and prompt-engineering workflows for intelligent document and chatbot systems.
- Worked on data preprocessing, feature engineering, model evaluation, and deployment pipelines for scalable ML applications.
- Implemented API integrations and backend services using FastAPI and LangChain-based orchestration workflows.
- Collaborated with cross-functional teams to deliver cloud-enabled AI solutions aligned with business requirements.
- Contributed to testing, debugging, monitoring, and optimization of production AI systems and data workflows.

Core Responsibilities

- Designed and deployed GenAI solutions using RAG architectures with grounded retrieval, citations, and fallback handling.
- Built agentic workflows with tool orchestration, planning, routing, and verification for multi-stage business tasks.
- Engineered prompts for reliable JSON outputs and implemented guardrails for safer LLM responses.
- Developed ML classification models with validation pipelines, leakage prevention, and threshold optimization.
- Implemented MLflow-based experiment tracking and reusable training and scoring pipelines.
- Created monitoring and data quality checks for drift, missing values, and performance degradation.
- Collaborated with business stakeholders to define measurable KPIs and acceptance criteria.
- Communicated insights and AI outputs through concise reporting and dashboards.

Projects

GenAI Patient Support Knowledge Assistant (RAG)

Healthcare Domain

GenAI / NLP Engineer

- Developed a Retrieval-Augmented Generation (RAG) chatbot using Azure OpenAI, LangChain, and Azure AI Search for healthcare knowledge retrieval.
- Processed SOPs, policy documents, and clinical information using embedding-based semantic search pipelines.
- Implemented prompt engineering, grounded retrieval, and response guardrails to improve factual accuracy and reduce hallucinations.
- Built scalable document ingestion and vector indexing workflows for enterprise healthcare data.

- Improved response relevance and retrieval performance through chunking strategies and context optimization.

Agentic AI for Airline IRROPS Decision Support

Airlines Domain

Agentic AI Engineer

- Built an agentic AI workflow for airline disruption management, automated policy validation, and rebooking recommendation generation.
- Designed multi-step orchestration pipelines using LangGraph, tool-calling, and routing mechanisms.
- Generated structured JSON outputs for flight rebooking decisions and policy citation workflows.
- Implemented verification, fallback handling, and monitoring pipelines to improve workflow reliability and latency tracking.
- Integrated business rules and decision validation mechanisms for operational AI workflows.

Customer Churn Prediction and Retention Analytics

Retail Domain

Data Scientist

- Developed customer churn prediction models using XGBoost and LightGBM for retention analytics and customer risk segmentation.
- Performed feature engineering, data preprocessing, and imbalance handling to improve model performance.
- Applied SHAP-based explainability techniques to identify key churn drivers and support business decision-making.
- Implemented MLflow experiment tracking and SQL-based reporting pipelines for model monitoring and reproducibility.
- Optimized classification thresholds using business KPIs to improve retention campaign effectiveness.

Education

Bachelour Degree in Computer Science

2023

Course Completed